PRINCE NZAMUWE

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EDUCATION

United States International University - Africa (USIU-Africa) — Nairobi, Kenya

BSc in Data Science & Analytics (Trimester Accelerated) — Expected 2028 (3rd Year)

- **Mastercard Foundation Scholar:** Full academic scholarship awarded for leadership, academic excellence, and community impact.
- **Coursework:** Machine Learning, Statistical Modeling & Inference, Data Structures & Algorithms, Data Warehousing & Mining, Database Systems.

ALX Africa — Remote

Software Engineering Programme (Foundations Phase) — 2023 – 2024

- Completed 9-month intensive systems engineering phase: C, Python, memory management, data structures, relational databases, Git team workflows.

OPEN SOURCE CONTRIBUTIONS

vega/altair (Vega-Lite Python visualization library — 10k+ stars) — PR #4202

- Implemented input validation and clear exception messaging for sequence length constraints in `selection_interval`; updated and clarified interval boundary documentation (resolving Issue #3896).

arnaudmiribel/streamlit-extras (Streamlit ecosystem — 1k stars) — PR #420 · PR #419

- Added directory ordering parameters to `directory_tree` to preserve filesystem insertion order; resolved cross-platform Windows executable handling in project build scripts.

FEATURED TECHNICAL PROJECTS

Credio — Credit Risk Scoring Pipeline & Application — Live Demo · Code

Python, Scikit-learn, Pandas, Streamlit, Pytest, Docker, GitHub Actions CI

- Engineered a leakage-controlled ML pipeline on 150,000 borrower records, resolving high missing-income rates and delinquency codes inside Scikit-learn estimators without test-set contamination.
- Benchmarked models under class imbalance (~6.7% default rate): baseline Logistic Regression (0.823 validation AUC) vs. Gradient Boosting (**0.869 test ROC-AUC**, **0.407 PR-AUC** on 30,000 held-out records).
- Configured calibrated risk tiers (1.1% to 39.0% observed defaults) and cost-based approval cutoffs; deployed Streamlit application with per-borrower explanations and 19 CI unit tests.

FinAccess Kenya — Financial Inclusion Analytics & Streaming Pipeline — Case Study · Code

Power BI (DAX), Python, Pandas, Star Schema Data Modeling

- Built a Python streaming pipeline converting a 290 MB Central Bank of Kenya / FSD household survey into an optimized 8 MB star-schema model (19,750 survey respondents, weighted to 28.1M adults).
- Developed an interactive 5-page Power BI dashboard reconciling weighted metrics against published national benchmarks (84.8% formal access, 9.9% complete exclusion) and mapping regional access gaps (48.5% to 94.0%).

PROFESSIONAL EXPERIENCE

Andela — Kigali, Rwanda — Jan 2024 – Sep 2024

Apprentice, Andela Technical Leadership Program (ATLP)

- Collaborated in a 9-person Agile engineering squad delivering full-stack features using TypeScript, React, and PostgreSQL; participated in code reviews and bi-weekly sprint retrospectives.
- Designed and normalized PostgreSQL database schemas for user management, product inventories, and order transaction pipelines.

Ida Technology — Kigali, Rwanda — Dec 2023 – Jun 2024

Software Engineering Trainee (Part-Time, NEET Program)

- Implemented modular REST backend endpoints in Django REST Framework with JWT authentication and role-based access.
- Translated Figma UI mockups into reusable, mobile-responsive React components with verified cross-browser compatibility.

TECHNICAL SKILLS & CERTIFICATIONS

Languages & Core:	Python, SQL (PostgreSQL, MySQL), TypeScript, JavaScript, R, MATLAB, HTML5/CSS3.
Data & Machine Learning:	Pandas, Scikit-learn, NumPy, Feature Engineering, Model Calibration, ROC-AUC / PR-AUC, Data Cleaning.
Frameworks & Tools:	FastAPI, Flask, Django REST Framework, React, Streamlit, Power BI (DAX), Docker, Git, Pytest, GitHub Actions.
Certifications & Honors:	Meta Back-End Developer · IBM Artificial Intelligence Fundamentals · freeCodeCamp JS Algorithms · MLH Hack Week Participant.